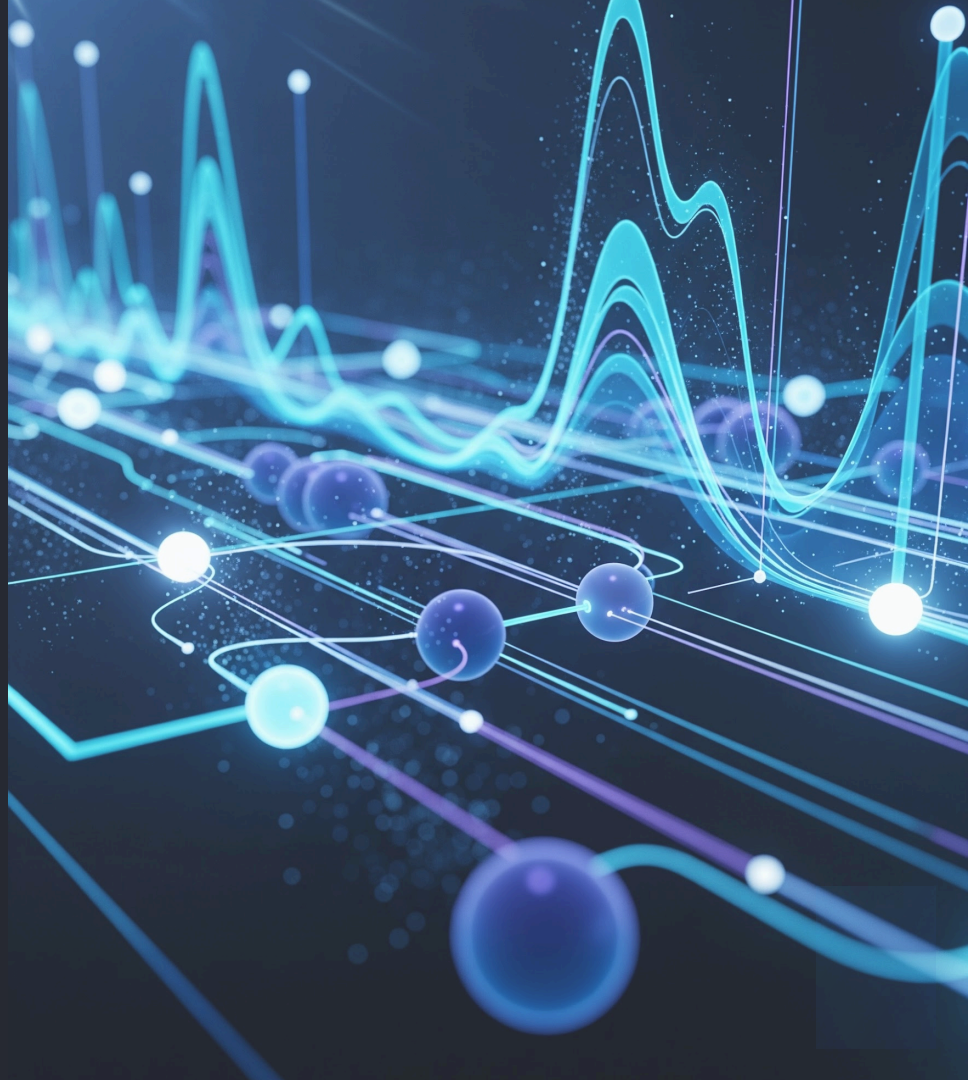


LINGUAE-ML SEMINAR

Machine Learning on *Temporal Graphs*

From Foundations to Emerging Frontiers

• 19/05/2026



A Bit About Myself



Bachelor and Master in **Computer Science** at Imperial College and Cambridge (2015–2019)



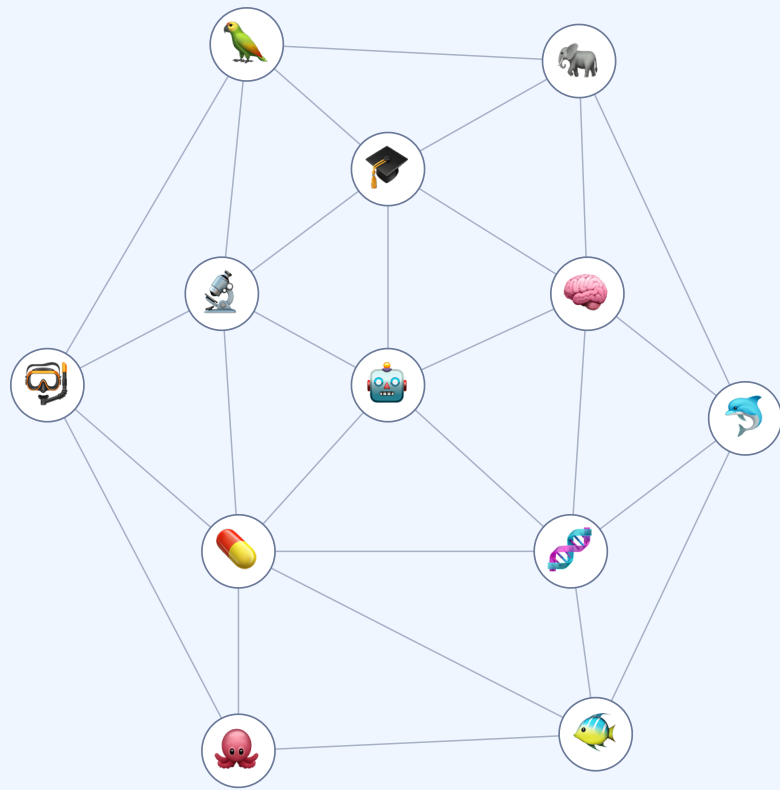
PhD on Machine Learning on Graphs between Twitter and Imperial College (2019–2023)



ML Researcher on **generative models for structural biology and drug discovery** at Vant AI (2024–2025)



PostDoc on **ML for decoding non-human communication** @ Sapienza

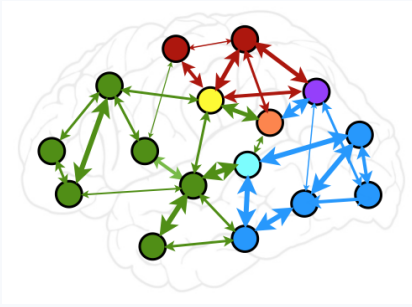


SECTION 01

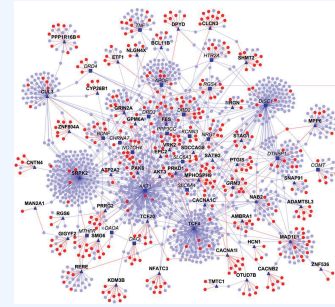
Why Should We Care About ML on Graphs?

Networks are everywhere

And graphs are a great way to model them



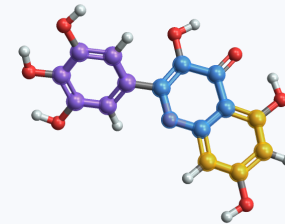
Functional Networks



Interaction Networks



Social Networks



Molecules

And graphs are a great way to model them



What model should we use for graphs?

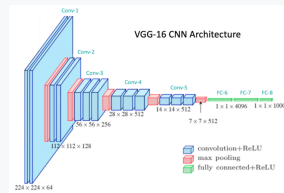
MODALITY

DATA

ARCHITECTURE

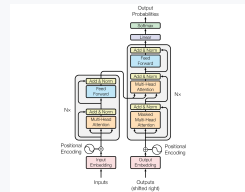


Images



Text

The cat sat on the mat



Graphs



?

SECTION 02

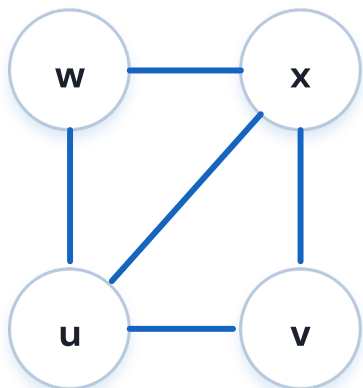
(Static) Graphs and Graph Tasks

Graphs: nodes, edges, and features

$$G = (V, E)$$

V = nodes

E = edges



$$A \in \mathbb{R}^{n \times n}$$

A = Adjacency Matrix

	u	v	w	x
u	0	1	1	1
v	1	0	0	1
w	1	0	0	1
x	1	1	1	0

$$X \in \mathbb{R}^{n \times d}$$

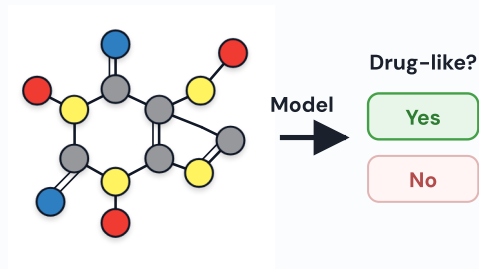
X = node features

	f1	f2	f3
u	0.8	0.4	1.7
v	0.2	0.9	1.4
w	0.6	0.1	1.9
x	0.4	0.7	1.6

Tasks on Graphs

Graph classification

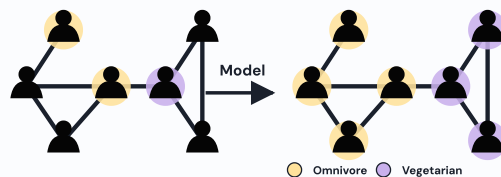
Predict a label for the whole graph.



Example: molecular property prediction, fraud detection, document classification.

Node classification

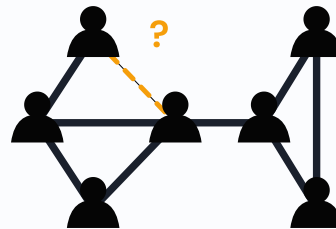
Predict labels for individual nodes using graph context.



Example: user attributes, protein function, paper topic prediction.

Link prediction

Score missing or future edges between pairs of nodes.



Example: recommendations, knowledge graph completion, social tie prediction.

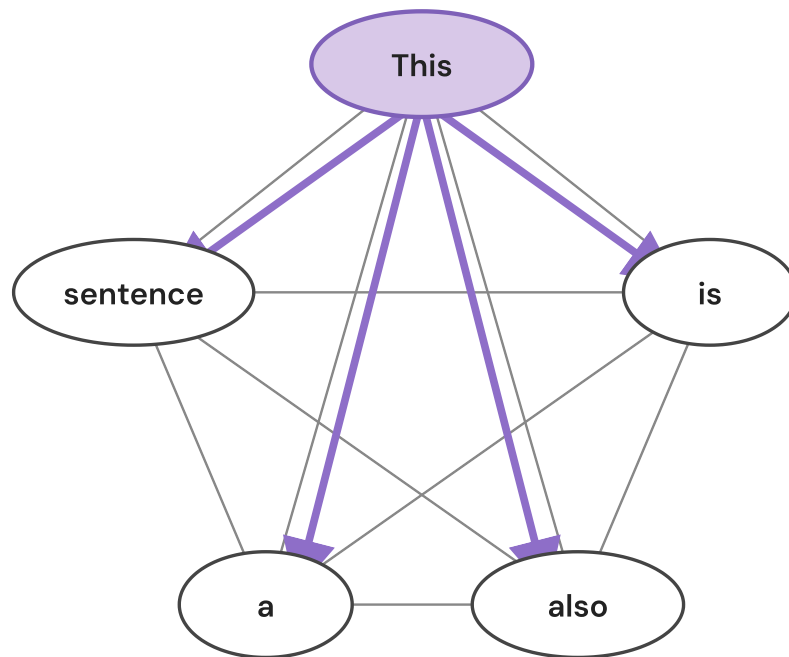
SECTION 03

(Static) Graph Neural Networks

Graph Neural Networks (GNNs)

Transformers are GNNs

On the fully connected graph



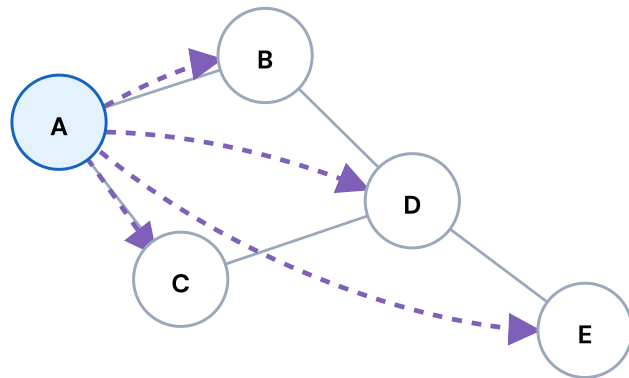
Graph Transformers [10]

Transformer-style attention, but with graph structure injected

CORE IDEA

- Self-attention lets every node aggregate from all other nodes
- Graph information enters through special positional encodings

$$\text{Attn}(i, j) \leftarrow q_i^\top k_j + b_{\text{graph}}(i, j)$$

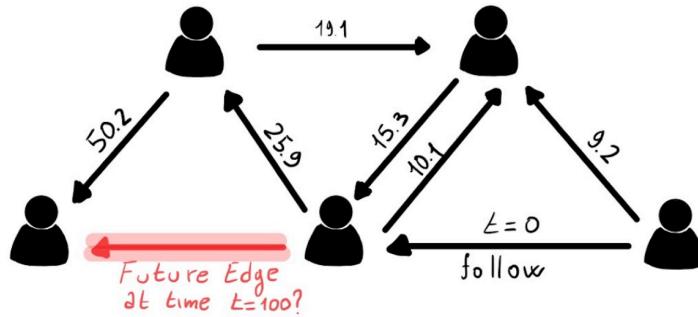


SECTION 04

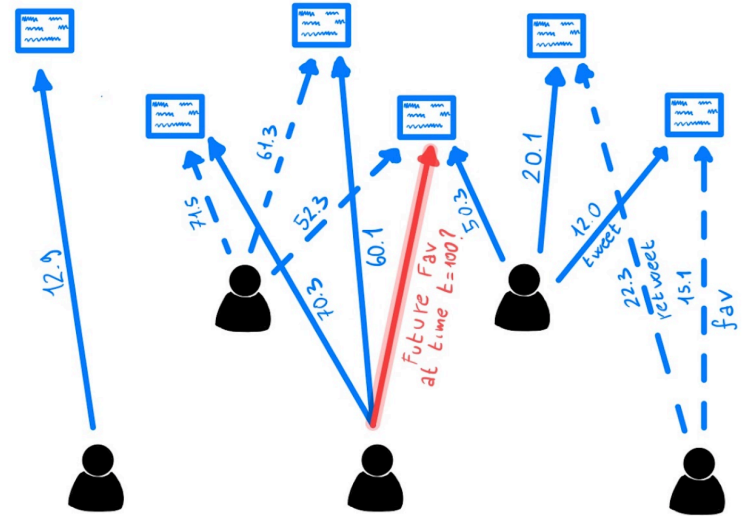
Dynamic Graphs

Some Examples of Dynamic Graphs

Graphs changing over time



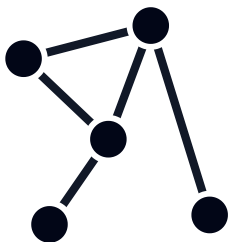
Social Networks



Interaction Networks

From Static to Dynamic Graphs

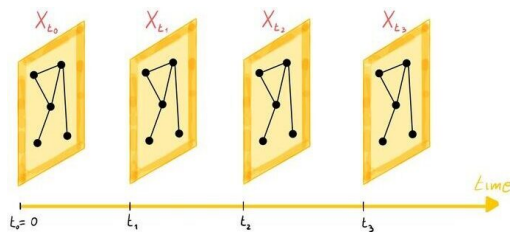
$$G = (V, E, X)$$



Static Graph

- No notion of time
- Examples: molecules, syntax trees

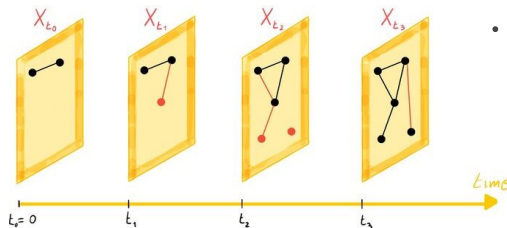
$$G_t = (V, E, X_t)$$



Spatio-Temporal Graph

- Fixed topology; changing features
- Regular time intervals
- Examples: traffic, weather sensors

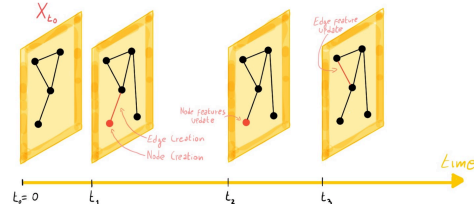
$$G_t = (V_t, E_t, X_t)$$



Discrete-Time DG (DTDG)

- Changing topology and features
- Regular time intervals → sequence of snapshots
- Examples: weekly trade networks

$$G(t) = \{x_{t_1}, x_{t_2}, \dots\} \quad t_1 \leq t_2 \leq \dots$$



Continuous-Time DG (CTDG)

- Most general formulation
- Sequence of timestamped events
- Examples: social, financial

Less General

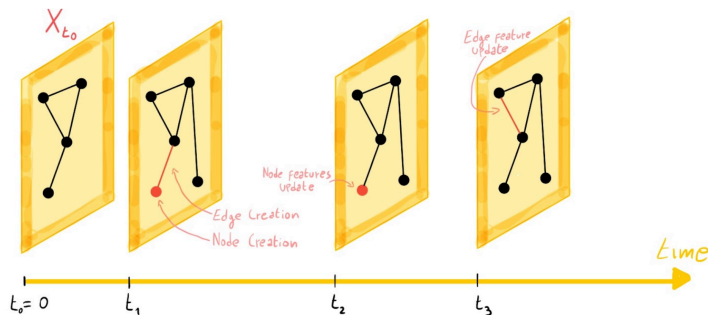


More General

Why Learning on Dynamic Graphs is Different

Temporal graph models must exploit the event history without collapsing it into one static snapshot

WHAT CHANGES



- Events arrive in order: creation, deletion, update, interaction
- The model should use both *what* happened and *when* it happened
- New tasks ask what happens next, or when it happens

WHY STATIC GNNS ARE NOT ENOUGH

event history
 $G(t \leq T)$



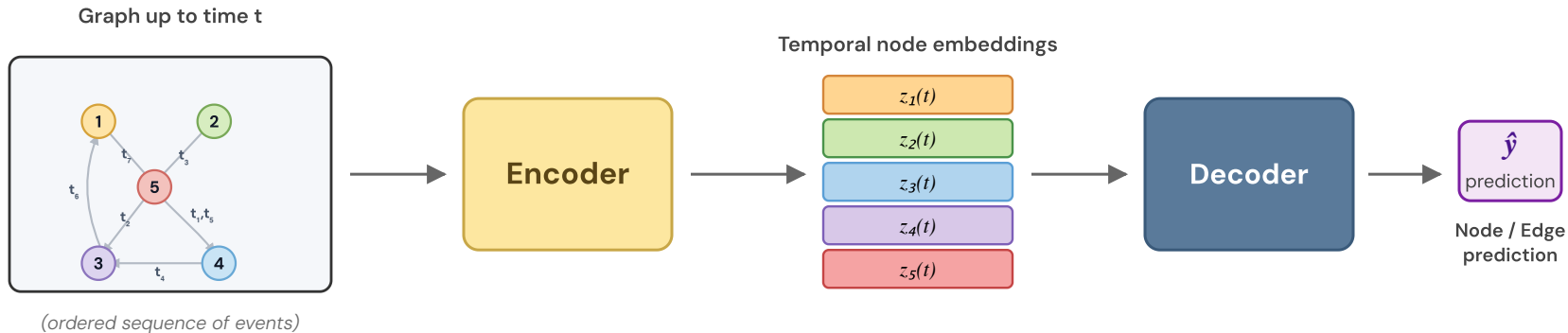
latest snapshot only

- *Information loss*: the last snapshot hides the evolution path
- *Inefficiency*: every new event can force repeated computation
- *No timing prediction*: static GNNs do not support predicting when something will happen

SECTION 05

Models

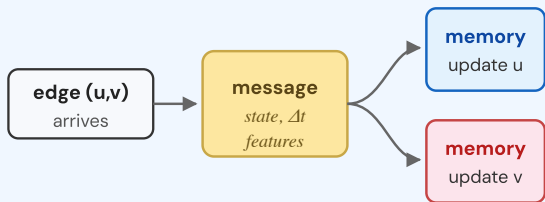
Temporal Graph Model



Memory-based Models [11, 12]

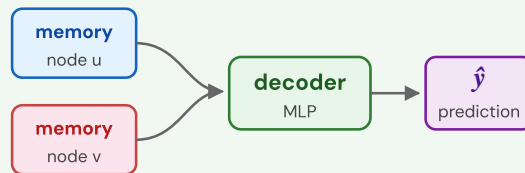
- Process events in order using an RNN, with a different hidden state per node
- Hidden state (memory) is a compressed representation of all past interactions of a node
- Memory is directly used as the temporal node embeddings

UPDATE TIME



Edge (u,v) builds a message from partner state, features & Δt ; only the two touched nodes update their memory.

PREDICT TIME



Memories serve directly as node embeddings, fed into decoder with no extra computation at query time.

Memory-based Models: Pros and Cons

A strong online model, but not yet a full temporal graph encoder

PROS

- Strong **sequentiality inductive bias**
- Cheap **online updates** after each new event

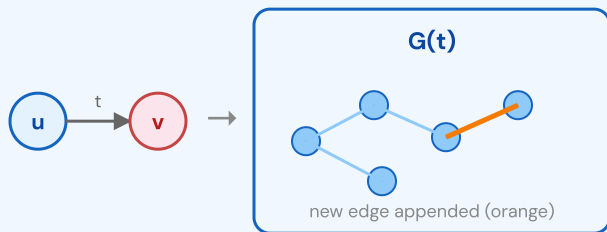
CONS

- Inactive nodes can become *stale*
- Graph context is mostly **local or indirect**
- Forced to process previous edges in sequential order

Graph-based Models [13]

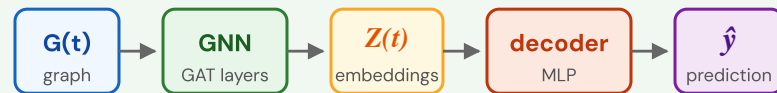
GNN on the graph of previous interactions, with timestamps as edge features

UPDATE TIME



Edge (u,v,t) is simply appended to $G(t)$ — no computation performed at event time.

PREDICT TIME



GNN runs on full $G(t)$ to produce node embeddings; must re-run for every new query — expensive at inference.

Graph-based Models: Pros and Cons [13]

Explicit graph structure mitigates staleness, but the GNN must re-run on every query

PROS

- No need for sequential processing in training
- Using the **graph explicitly** → mitigates staleness problem

CONS

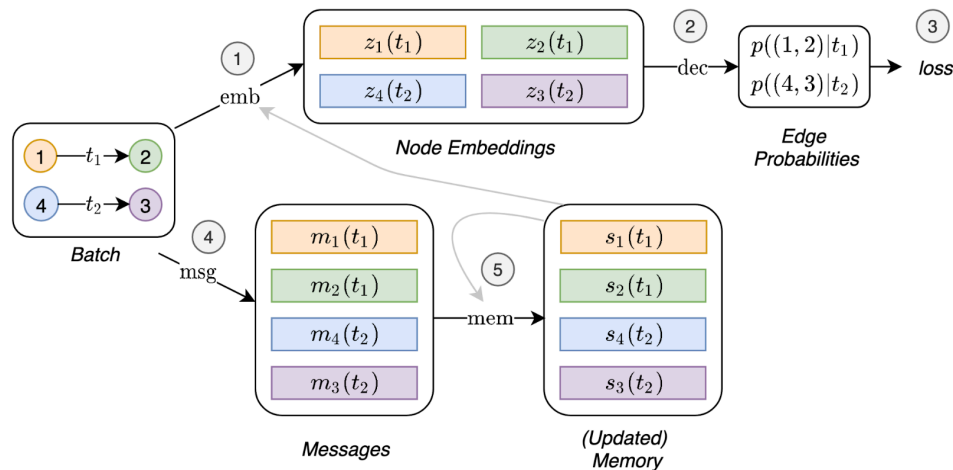
- Can only handle *edge addition* events
- Need to re-run GNN after each new event → **inefficient at inference**

TGN: Temporal Graph Networks [14]

Our work: a modular framework that combines memory-based and graph-based temporal learning

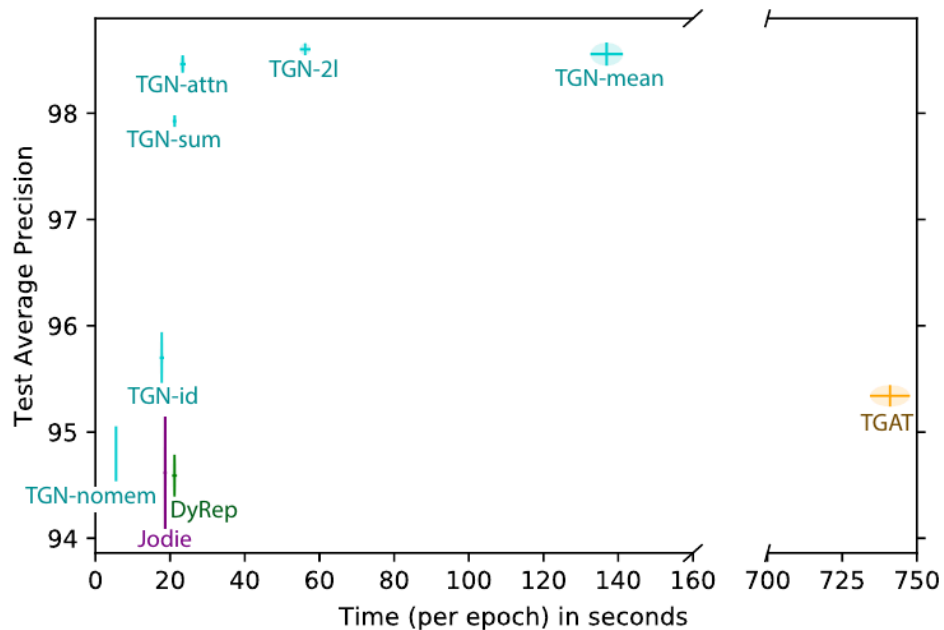
TGN CONTRIBUTION

- General framework combining the best of memory-based and graph-based approaches
- Updates node memories from events, then uses a GNN over the interaction graph to produce embeddings
- Generalizes DyRep, Jodie, and TGAT in one notation



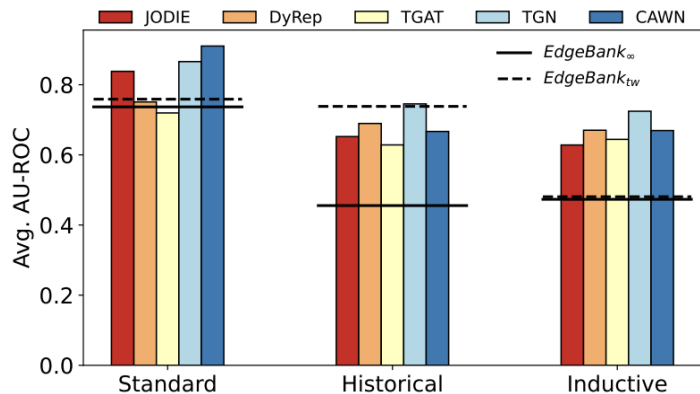
TGN: State-of-the-Art in 2020

High accuracy with much lower per-epoch cost than the previous methods



TGN: Still a Strong Baseline Today

Years of follow-up work, yet TGN remains hard to beat on standard benchmarks [14, 15]



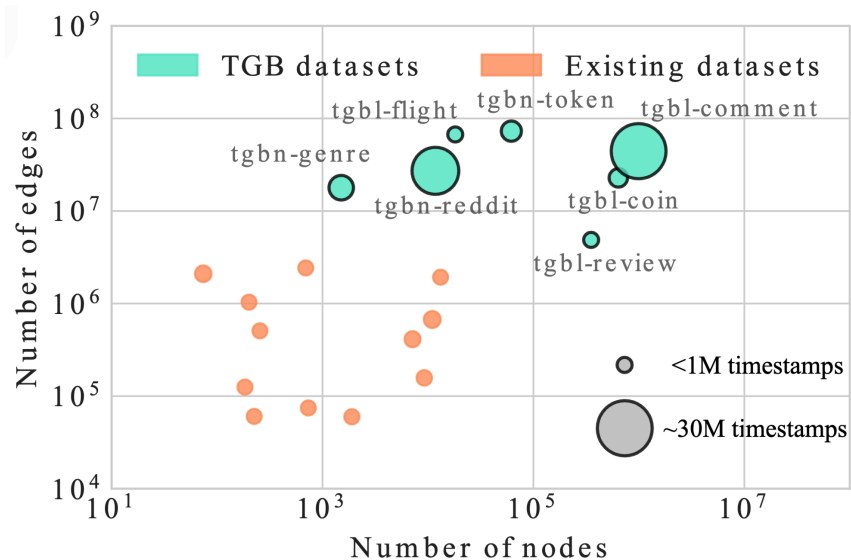
Datasets	Flickr	YouTube	Patent	WikiLink
JODIE	46.21 ± 0.83	41.67 ± 2.86	24.60 ± 0.38	57.94 ± 1.33
DyRep	38.04 ± 4.19	35.12 ± 4.13	21.01 ± 1.14	42.63 ± 1.33
TGAT	23.53 ± 3.35	43.56 ± 2.53	8.49 ± 0.18	OOT
TGN	46.03 ± 6.78	55.16 ± 5.89	22.83 ± 2.25	62.94 ± 2.16
CAWN	48.69 ± 6.08	47.55 ± 1.08	12.34 ± 0.47	OOT
TCL	40.00 ± 1.76	50.17 ± 1.98	10.60 ± 1.75	43.02 ± 2.16
GraphMixer	45.01 ± 0.08	58.87 ± 0.12	18.97 ± 2.54	48.57 ± 0.02
DyGFormer	49.58 ± 2.87	46.08 ± 3.44	14.20 ± 2.93	OOT

SECTION 06

Benchmarks

Temporal Graph Benchmark [4]

Diverse, large datasets and unified evaluation



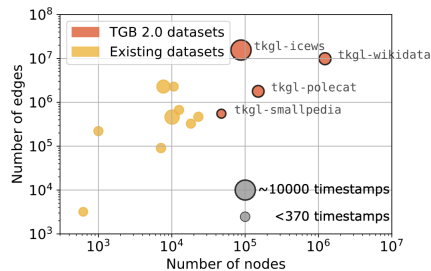
KEY FINDINGS

- Previous datasets were too easy and saturated
- Exposed that simple historical baselines can be surprisingly strong
- Large new datasets make scalability part of the benchmark

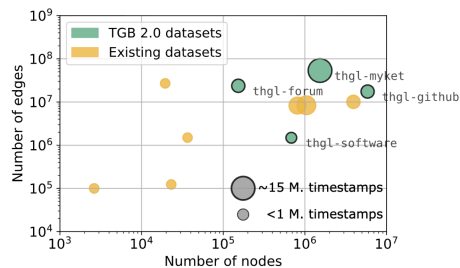
9 datasets · 5 domains · up to 72M edges

Temporal Graph Benchmark 2.0 [5]

TGB but for knowledge graphs



(a) Novel Temporal Knowledge Graphs



(b) Novel Temporal Heterogeneous Graphs

KEY FINDINGS

- Edge and relation type information is crucial for strong performance
- Simple heuristic baselines remain competitive with more complex methods
- Many methods fail to run on the largest datasets, making scalability a central result

8 datasets · 5 domains · up to 53M edges

SECTION 07

Unifying Models for Discrete and Continuous Time Graphs

UTG: Unifying Snapshot & Event-Based Models

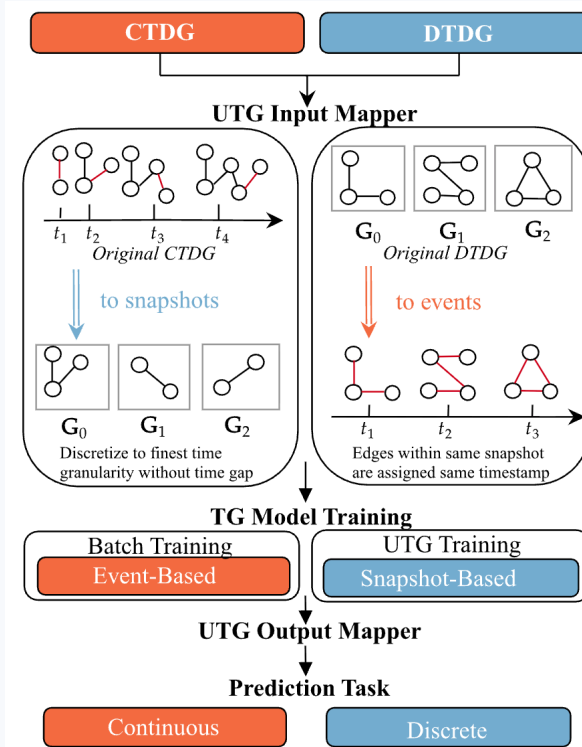
Huang, Poursafaei, Rabbany, Rabusseau, Rossi, LoG 2024

THE PROBLEM

Snapshot (DTDG) and event-based (CTDG) models developed **in isolation**: limited cross-comparison and no unified evaluation.

CONTRIBUTION

- **Input mapper**: convert CTDG to snapshots and DTDG to events, so any model can run on any data
- **UTG training**: use streaming training to make snapshot-based models operate on event streams
- **Output mapper**: align predictions to either continuous-time or discrete-time tasks



UTG: Key Findings

SPEED

Snapshot-based models are **$\geq 10\times$ faster** at inference than most event-based models.

PERFORMANCE

With UTG training, snapshot-based models **match TGN & GraphMixer** even on event-based (CTDG) datasets.

INSIGHT

NAT & DyGFormer's edge comes from **joint neighbourhood features**, not from the event-based format: these can be added to snapshot models too.

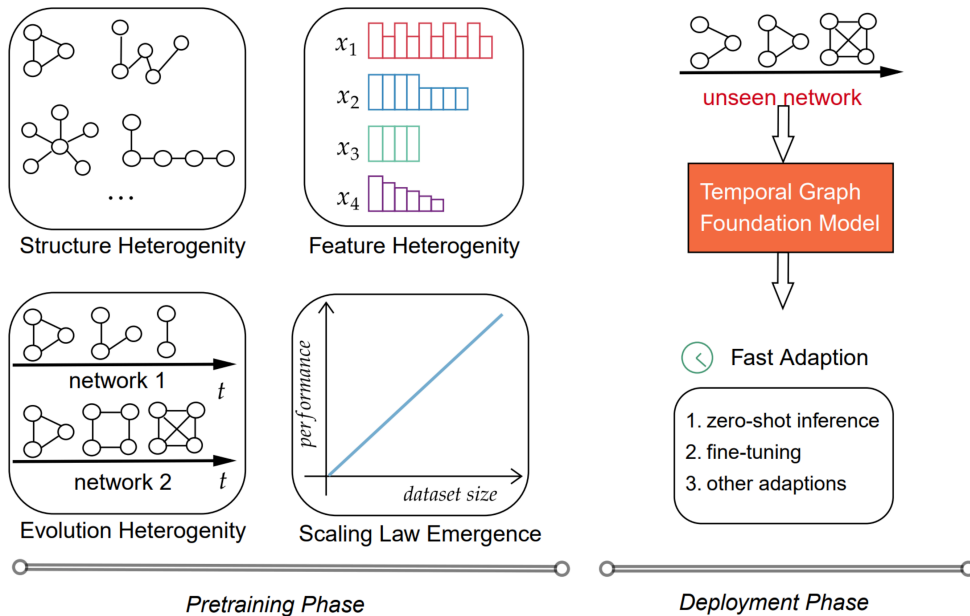
SECTION 08

Next Frontiers

Foundation Models for TG

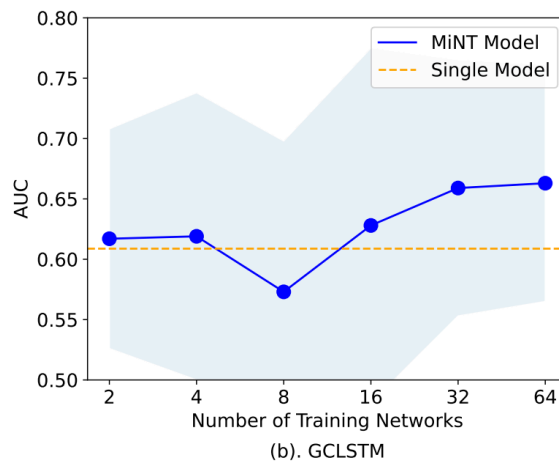
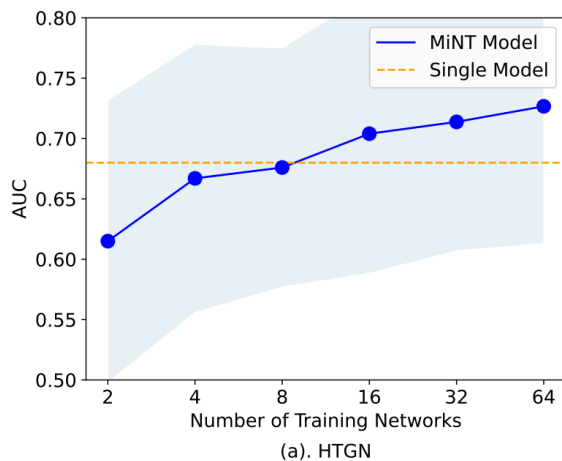
Existing TG models are **trained and tested on the same graph**.

Can we train a single TG model that works on *unseen graphs*, ideally from different domains?



MiNT: Multi-Network Training for Transfer Learning on Temporal Graphs

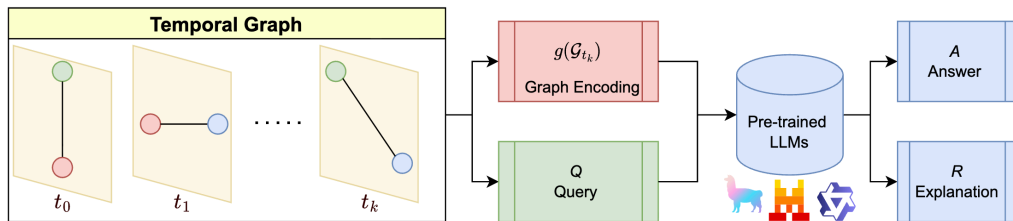
- **New dataset:** 84 distinct ERC-20 token transaction networks
- **New framework** to train existing TG models on multiple graphs simultaneously
- **Scaling law for TGs:** model performance improves as it is trained on more networks



Can we use LMs directly on temporal graphs?

TGTalker

Translate temporal graph structure into natural language and feed it to a **pre-trained LLM** — no fine-tuning required.



Four prompt components

- **Background set** — recent edges as context
- **Example set** — 5-shot Q&A pairs
- **Query set** — target edge to predict
- **Temporal neighbors** — 1-hop recent neighbours

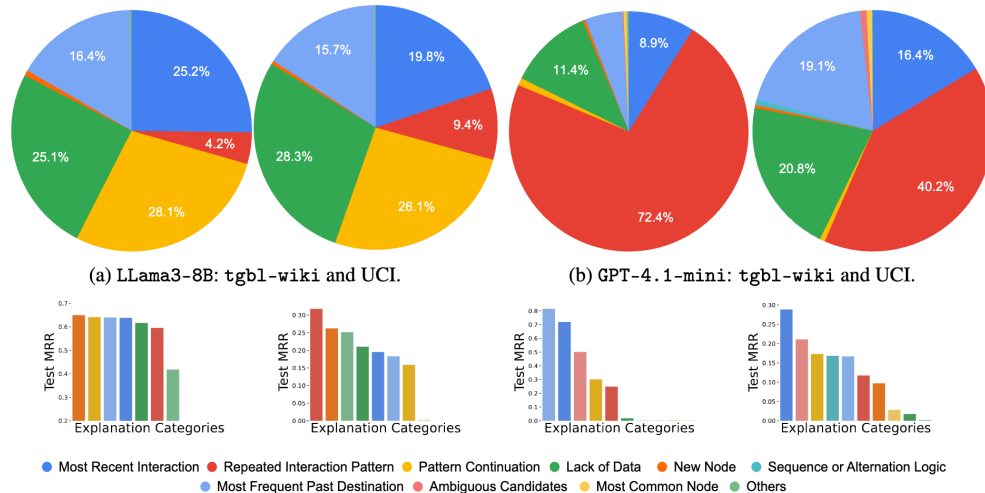
TGTalker: Results & Explainability

RESULTS

- Competitive without fine-tuning
- Consistently outperforms TGN and HTGN

10 EXPLANATION CATEGORIES DISCOVERED

- Most Recent Interaction** $\approx$ EdgeBank heuristic
- Most Frequent Destination** $\approx$ PopTrack heuristic
- Novel patterns: sequence logic, analogy-based inference



Resources for TGL

Blog posts, talks, surveys, libraries & datasets

BLOG POSTS

[Temporal Graph Learning in 2023](#)

[Temporal Graph Learning in 2024](#)

TALKS & READING GROUPS

[Andy Huang Lecture Series](#)

[Temporal Graph Reading Group](#)

recent papers and talks

SURVEY

[Graph Neural Networks for temporal graphs: state of the art, open challenges, and opportunities](#)

LIBRARIES

[TGM](#)

Temporal Graph Models library

DATASETS

[TGB](#)

Temporal Graph Benchmark

[TGB-Seq](#)

Sequential Temporal Graph Benchmark

(a small taste of ongoing work)

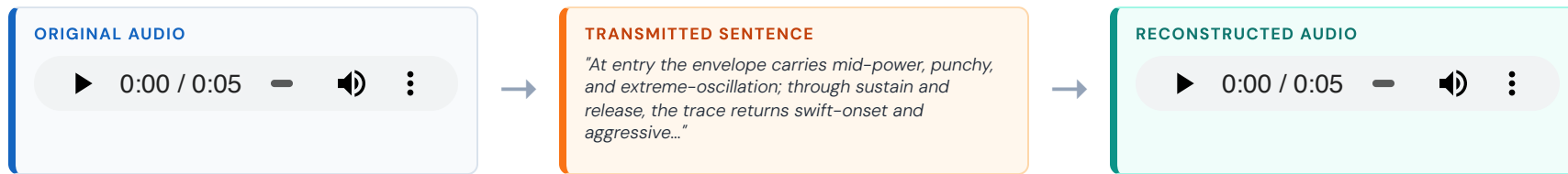
SECTION 09

My Current Directions



Communicating Sound Through Natural Language

A sender agent describes a sound; a receiver agent decodes the description back to audio



COMPARISON

[illegible]

Communicating Sound Through Natural Language

A sender agent describes a sound; a receiver agent decodes the description back to audio

ORIGINAL AUDIO



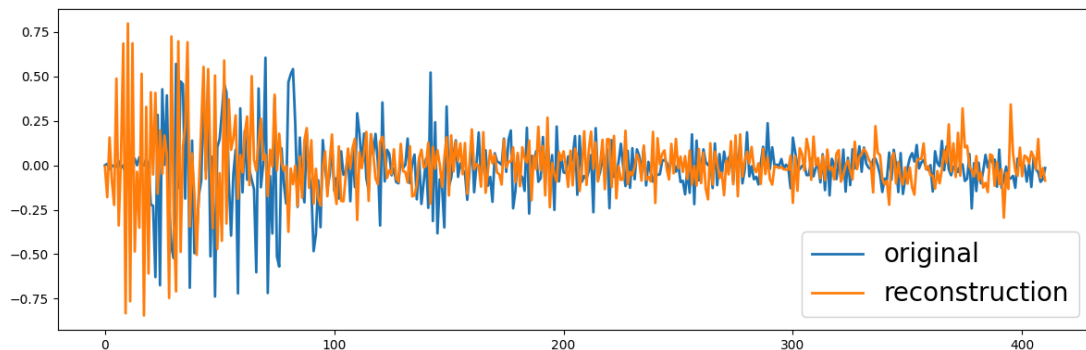
TRANSMITTED SENTENCE

"At entry the envelope carries mid-power, punchy, and extreme-oscillation; through sustain and release, the trace returns swift-onset and aggressive..."

RECONSTRUCTED AUDIO



WAVEFORMS



OpenWhistle Dataset

- Very rich and unique **dolphin dataset** collected over 5 years
- New benchmark for *whistle detection and classification*

Dataset	# Whistles	Voc. hours	Time span (yrs)	Stable pod (# indiv.)	Setting	Seq. context	Open
OpenWhistle Pretraining	~180,000*	114.3	5.0	● (5)	Semi-nat.	●	●
OpenWhistle Expert subset	8,354	1.9	0.42	● (5)	Semi-nat.	○	●
DOLPHINFREE	4,600	7.3	2.0	○	Wild	○	●
Di Nardo et al., 2025	3,111	0.6	0.003	● (7)	Captive	○	●
Watkins MMSD	566	N/R	70+	○	Wild	○	●
Korkmaz et al., 2023	~29,000*	6.8	0.07	○	Semi-nat.	●	🟡
Sicily Strait PAM	14,048	N/R	1.2	○	Wild	●	⊗
DCLDE 2011	6,011	0.7	4.0	○	Wild	○	⊗
SDWD	N/R	N/R	43+	● (293)	Wild (C&R)	○	🟡

* Estimated from total vocalization duration and mean whistle duration.

THANK YOU

Questions?

ML on Temporal Graphs · Linguae-ML Seminar · 19/05/2026